

**Phys 247 : Fall 2026**

Problem Set 3 (v2)

Assigned 9/18, Due Monday 9/28

Submit HW by Start of Class, No Late Grades  
Start early!

**Problem 1** (easy)

KK 3.19 - Compound Springs

KK 3.7 - Tension, Gravity, and Friction

**Problem 2** (medium)

KK 3.8 - Block and Wedge, now with Friction

**Problem 3** (hard)

KK 3.12 - Sailboat Physics

Hint 1: Can you motivate  $dT/d\theta = -\mu T$ ?

Hint 2: Carefully review example 3.7.

**Problem 4** (easy)

KK 4.3 - Water Molecule

KK 4.1 - Non-uniform rod

**Extra Credit** (medium)

KK 4.15 - Cart Jumpers

**Submission Instructions:**

Scan a PDF. Either use an office machine or make clear photos with your cell phone. If you make photos, please collect images into PDF.

Upload to Gradescope on canvas. Do **not** use AI to enhance, check, or auto-latex your solutions. You're encouraged to work together in groups, but everyone must write and submit solutions individually.